[No. of Printed Pages – 4] **1921**

IT201 Enrol. No. ...903.....

[ET]

END SEMESTER EXAMINATION : APRIL-MAY, 2025

JAVA PROGRAMMING

Time : 3 Hrs. *Maximum Marks : 60*

Note: ***Attempt questions from all sections as directed.***

SECTION – A (24 Marks)

Attempt any **four** *questions out of* **five.**

Each question carries **06** *marks.*

1. Describe the concept of method overriding. How does it differ from method overloading in Java?

2. Evaluate the advantages and disadvantages of placing all classes in a single package versus organizing them into multiple packages. When would you use one approach over the other?

P.T.O.

3. Design a Java program that creates two threads by extending the Thread class. One thread should print numbers from 1 to 10, and the other should print letters from A to J. Ensure that both threads run concurrently.

4. How does the use of generics prevent ClassCastException during runtime? Write a simple user-defined generic class Box that can store an object of any type. The class should have methods to set and get the value of the stored object.

5. What is the difference between Central Repository, Local Repository, and Remote Repository in Maven, and how do they interact in dependency resolution?

SECTION – B (20 Marks)

Attempt any **two** *questions out of* **three.**

Each question carries **10** *marks.*

6. (a) Explain the concept of Lambda Expression and it's importance in Java programming. Also write a Java program to implement a lambda expression to calculate the sum of squares of all odd and even numbers in a list. (5)

(b) Differentiate between Errors and Exceptions. Write a Java program to create a method that takes a string as input and throws an exception if the string does not contain vowels. (5)

7. (a) Describe how InputStream and OutputStream differ from Reader and writer classes in terms of functionality. Write a Java program that reads data from a binary file and writes it to another binary file using byte streams (FileInputStream and FileOutputStream). (5)

(b) Write a program that demonstrates the use of ArrayList and LinkedList to store and retrieve data.What are the advantages and disadvantages of using each? (5)

8. (a) Create a custom annotation that uses meta-annotations (@Retention, @Target, @Documented) and apply it to a method. (5)

(b) Create an abstract class Shape with a method calculateArea(). Implement two subclasses (Circle and Rectangle), override calculateArea(), and demonstrate dynamic method dispatch. (5)

P.T.O.

SECTION – C (16 Marks)

(Compulsory)

9. (a) Create a quiz application with the use of Java collection framework. The application should be able to perform tasks such as store the list of questions and answers with the use of approriate framework collection. The quiz should be condcuted and store the respones, keeping track of the scores and display the results after completion of the quiz. Use of Map, List or Sets may be done wherever needed. (8)

(b) How Maven resolves dependency conflicts when two different dependencies transitively introduce the same artifact with different versions.

Develop a banking system where multiple threads can deposit and withdraw money from a shared bank account, ensuring **thread safety using synchronized blocks.** (8)